starCAT MICROGLIA_V1 — cross-validation

22 human-microglia datasets · generated 2026-05-25 19:35 UTC

1. Overview

This report compares starCAT `MICROGLIA_V1` annotations against author-defined labels and state-scores that already exist in each dataset's obs table. The catalog assigns each cell a soft mixture across 21 microglia / DC / macrophage programs and a single argmax `Identity_Label` from 13 identity categories.

For each dataset we show:

1. **QA flags** — gene overlap with the catalog (out of 2,727), median per-cell residual (high = poor catalog fit), and the WholeCell-baseline / DC-soft-signal fractions documented in `ANNOTATE_CELLS.md` §7.5.
2. **Identity_Label distribution** — the argmax assignment per cell.
3. **Cross-validation** — where the author's obs columns include state scores (`HM/DAM/IRM/HLA/CRM/RM`, `*_enriched1`, etc.) or cell-type / cluster labels, we compute mean author-score per starcat label or a confusion-matrix-style cross-tab.

Strong agreement on rows where starcat says "DAM" and author scores show high DAM is the simplest sanity check that **both** annotations describe the same biology.

1.1. Summary table

Dataset	Tech	Cells	Resid	WC%	DC%	Dominant label
Schirmer2019/ schirmer2019_microglia	snRNA-seq	1,425	1.700	0.0%	29.1%	Homeostatic (60%)
Pan2024/Microglia	10x_chromium	1,621	1.872	0.0%	37.8%	Homeostatic (44%)
Smajic2022/ smajic2022_microglia	snRNA-seq	3,903	1.183	0.0%	42.9%	Homeostatic (71%)
LuChen2025/101010sn1_control	10x_chromium	6,033	1.547	0.4%	57.7%	Homeostatic (50%)
HouZielonka2023/ LPS_sema_micro_invitro	10x_chromium	8,395	2.885	0.0%	99.6%	DC_pDC (39%)
HouZielonka2023/ noLPS_sema_micro_invitro	10x_chromium	8,611	2.543	0.8%	99.7%	Lipid_LAM (54%)
Olah2020/ olah2020_microglia	10x_chromium	16,245	4.099	11.8%	56.3%	HLA_MHCII (34%)
AlbertiniZielonka2025/ scrna_lecMicro_cleaned_integrated_sctransform	10x_chromium	22,420	1.972	47.4% ⚠ likely whole-cell	53.7%	Homeostatic_MERTK (47%)
Gabitto2024/ Seattle_MTG_MG	10x_chromium	40,000	1.090	0.8%	64.7%	Homeostatic (56%)
Simpson2025/ module_scored_sorl1	10x_chromium	42,270	2.974	18.9%	44.7%	Homeostatic_MERTK (75%)
Simpson2025/ seurat_obj_sorl1_final	10x_chromium	42,270	1.663	16.8%	56.4%	Homeostatic_MERTK (77%)
Gabitto2024/ Seattle_PFC_MG	10x_chromium	42,486	1.095	0.7%	67.1%	Homeostatic (57%)
Davis2025/20240812_sex_microglia	10x_chromium	43,552	1.333	25.0%	49.9%	Homeostatic_MERTK (58%)
Davis2025/ davis2025.20250619	10x_chromium	43,552	1.333	25.0%	49.9%	Homeostatic_MERTK (58%)
Gazestani2023/ microglia	10x_chromium	67,513	1.226	0.3%	44.8%	Homeostatic (59%)
Blackburn2026/ seurat_obj_final	10x_chromium	73,888	2.408	17.8%	61.1%	Homeostatic_MERTK (74%)
Blackburn2026/ trisomy_seurat_obj_w_umap	10x_chromium	73,888	3.256	21.2%	54.0%	Homeostatic_MERTK (71%)

Dataset	Tech	Cells	Resid	WC%	DC%	Dominant label
Borrie2025/ LOF.sc.teset.counts	10x_chromium	77,684	2.353	39.8% ⚠ likely whole-cell	54.8%	Homeostatic_MERTK (53%)
GreenFujita2024/ microglia	10x_chromium	86,612	1.229	0.0%	63.4%	Homeostatic (63%)
Blackburn2026/ seurat_obj_trisomy_initial	10x_chromium	88,582	2.647	14.2%	72.1%	Homeostatic_MERTK (68%)
Mathys2024/ multiregion_microglia	10x_chromium	92,800	1.280	0.0%	50.4%	Homeostatic (65%)
Mancuso_2024/ Mancuso_2023	10x_chromium	127,755	1.954	56.0% ⚠ likely whole-cell	60.8%	Homeostatic_MERTK (47%)

2. Per-dataset cross-validation

2.1. Schirmer2019 / schirmer2019_microglia

Tech: snRNA-seq, n_cells: 1,425 ⚠ 43 pending columns not merged

2.1.1. QA flags

Cells	1,425
Gene overlap	2,727 / 2,727
Median residual	1.700
WholeCell% (high)	0.0%
DC% (> 0.05)	29.1% ⚠ DC contamination

2.1.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	851	59.7%
DAM	241	16.9%
Homeostatic_MERTK	146	10.2%
Lipid_LAM	72	5.1%
Border_CAM	49	3.4%
DC_pDC	19	1.3%
Cycling	18	1.3%
Monocyte_Derived	18	1.3%
DC_cDC2	5	0.4%
HLA_MHCII	3	0.2%
other (2)	3	0.2%

No comparable author state-scores or label columns detected in this dataset; starcat annotation stands on its own.

2.2. Pan2024 / Microglia

Tech: 10x_chromium, n_cells: 1,621 ⚠ 43 pending columns not merged

2.2.1. QA flags

Cells	1,621
Gene overlap	2,727 / 2,727
Median residual	1.872
WholeCell% (high)	0.0%
DC% (> 0.05)	37.8% ⚠ DC contamination

2.2.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	711	43.9%
Homeostatic_MERTK	256	15.8%
Lipid_LAM	220	13.6%
DAM	158	9.7%
Border_CAM	145	8.9%
Monocyte_Derived	61	3.8%
DC_pDC	27	1.7%
Cycling	16	1.0%
IRM	8	0.5%
DC_cDC2	7	0.4%
other (2)	12	0.7%

No comparable author state-scores or label columns detected in this dataset; starcat annotation stands on its own.

2.3. Smajic2022 / smajic2022_microglia

Tech: snRNA-seq, n_cells: 3,903 ⚠ 43 pending columns not merged

2.3.1. QA flags

Cells	3,903
Gene overlap	2,727 / 2,727
Median residual	1.183
WholeCell% (high)	0.0%
DC% (> 0.05)	42.9% ⚠ DC contamination

2.3.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	2,767	70.9%
DAM	538	13.8%
Lipid_LAM	149	3.8%
Homeostatic_MERTK	113	2.9%
Border_CAM	84	2.2%
DC_pDC	60	1.5%
HLA_MHCII	50	1.3%
DC_cDC2	37	0.9%
Monocyte_Derived	33	0.8%
Cycling	32	0.8%
other (3)	40	1.0%

No comparable author state-scores or label columns detected in this dataset; starcat annotation stands on its own.

2.4. LuChen2025 / 101010sn.1.micro

Tech: 10x_chromium, n_cells: 6,033

2.4.1. QA flags

Cells	6,033
Gene overlap	2,727 / 2,727
Median residual	1.547
WholeCell% (high)	0.4%
DC% (> 0.05)	57.7% ⚠ DC contamination

2.4.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	3,002	49.8%
Homeostatic_MERTK	1,179	19.5%
DAM	564	9.3%
Lipid_LAM	418	6.9%
Border_CAM	299	5.0%
DC_pDC	178	3.0%
Cycling	89	1.5%
HLA_MHCII	81	1.3%
Monocyte_Derived	78	1.3%
DC_cDC1	55	0.9%
other (3)	90	1.5%

2.4.3. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `first\_subtypes`. Concentration along rows = agreement.

starcat label	micro_0	micro_1	micro_2	micro_3	micro_4	micro_5	micro_6
Border_CAM	**126**	22	15	60	29	32	15
Cycling	31	9	8	14	20	4	3
DAM	**208**	29	**117**	50	88	45	27
DC_Mature	·	·	·	3	·	·	1
DC_cDC1	10	3	8	13	13	5	3
DC_cDC2	12	·	6	2	5	6	3
DC_pDC	58	22	17	40	20	18	3
HLA_MHCII	24	6	21	·	20	9	1
Homeostatic	**404**	**678**	**748**	**638**	**314**	**207**	13
Homeostatic_MERTK	**286**	**440**	71	**128**	**145**	**102**	7
IRM	10	4	21	2	8	5	2
Lipid_LAM	**173**	12	**113**	36	29	44	11

Author column: `first\_subtypes`

2.5. HouZielonka2023 / LPS_sema_micro_invitro

Tech: 10x_chromium, n_cells: 8,395 ⚠ 43 pending columns not merged

2.5.1. QA flags

Cells	8,395
Gene overlap	2,697 / 2,727
Median residual	2.885
WholeCell% (high)	0.0%
DC% (> 0.05)	99.6% ⚠ DC contamination

2.5.2. Identity_Label distribution

Identity_Label	n	%
DC_pDC	3,255	38.8%
Lipid_LAM	3,113	37.1%
DC_cDC2	1,149	13.7%
Monocyte_Derived	327	3.9%
DAM	192	2.3%
Border_CAM	154	1.8%
IRM	100	1.2%
Homeostatic	34	0.4%
Homeostatic_MERTK	23	0.3%
DC_cDC1	18	0.2%
other (2)	30	0.4%

2.5.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM
DC_pDC	3,255	0.22
Lipid_LAM	3,113	0.19
DC_cDC2	1,149	0.24
Monocyte_Derived	327	0.22
DAM	192	0.27
Border_CAM	154	0.31
IRM	100	0.23
Homeostatic	34	0.28
Homeostatic_MERTK	23	0.31
DC_cDC1	18	0.21

Author column mapping: HM→HM_enriched1

2.5.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `cluster`. Concentration along rows = agreement.

starcat label	0	1	2
Border_CAM	14	**136**	4
Cycling	2	13	·
DAM	21	**170**	1
DC_cDC1	9	9	·
DC_cDC2	**268**	**872**	9
DC_pDC	**1,216**	**2,025**	14
HLA_MHCII	·	15	·
Homeostatic	28	6	·
Homeostatic_MERTK	5	18	·
IRM	4	11	85
Lipid_LAM	**2,651**	**448**	14
Monocyte_Derived	**106**	**221**	·

Author column: `cluster`

2.6. HouZielonka2023 / noLPS_sema_micro_invitro

Tech: 10x_chromium, n_cells: 8,611 ⚠ 43 pending columns not merged

2.6.1. QA flags

Cells	8,611
Gene overlap	2,697 / 2,727
Median residual	2.543
WholeCell% (high)	0.8%
DC% (> 0.05)	99.7% ⚠ DC contamination

2.6.2. Identity_Label distribution

Identity_Label	n	%
Lipid_LAM	4,638	53.9%
DC_cDC2	1,372	15.9%
DC_pDC	1,256	14.6%
Border_CAM	414	4.8%
Homeostatic_MERTK	301	3.5%
Homeostatic	251	2.9%
DAM	125	1.5%
Monocyte_Derived	117	1.4%
IRM	54	0.6%
HLA_MHCII	54	0.6%
other (3)	29	0.3%

2.6.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM
Lipid_LAM	4,638	0.27
DC_cDC2	1,372	0.35
DC_pDC	1,256	0.32
Border_CAM	414	0.41
Homeostatic_MERTK	301	0.43
Homeostatic	251	0.38
DAM	125	0.33
Monocyte_Derived	117	0.26
IRM	54	0.34
HLA_MHCII	54	0.32

Author column mapping: HM→HM_enriched1

2.6.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `cluster`. Concentration along rows = agreement.

starcat label	0	1	2
Border_CAM	32	**382**	·
Cycling	2	7	·
DAM	21	**104**	·
DC_Mature	1	·	·
DC_cDC1	5	14	·
DC_cDC2	**377**	**994**	1
DC_pDC	**509**	**747**	·
HLA_MHCII	15	38	1
Homeostatic	**196**	55	·
Homeostatic_MERTK	**107**	**194**	·
IRM	10	9	35
Lipid_LAM	**3,829**	**802**	7

Author column: `cluster`

2.7. Olah2020 / olah2020_microglia

Tech: 10x_chromium, n_cells: 16,245 ⚠ 43 pending columns not merged

2.7.1. QA flags

Cells	16,245
Gene overlap	2,723 / 2,727
Median residual	4.099
WholeCell% (high)	11.8%
DC% (> 0.05)	56.3% ⚠ DC contamination

2.7.2. Identity_Label distribution

Identity_Label	n	%
HLA_MHCII	5,450	33.5%
Homeostatic	3,698	22.8%
Lipid_LAM	1,789	11.0%
Homeostatic_MERTK	1,005	6.2%
DC_cDC2	997	6.1%
DC_pDC	920	5.7%
IRM	710	4.4%
DAM	473	2.9%
Border_CAM	438	2.7%
DC_cDC1	320	2.0%
other (3)	445	2.7%

2.7.3. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `cell\_type`. Concentration along rows = agreement.

starcat label	microglial cell	T cell	neural cell	monocyte	erythrocyte	B cell
Border_CAM	**427**	1	10	·	·	·
Cycling	**167**	2	1	·	·	·
DAM	**473**	·	·	·	·	·
DC_Mature	56	·	·	·	·	·
DC_cDC1	**313**	1	1	·	5	·
DC_cDC2	**994**	1	1	·	1	·
DC_pDC	**898**	8	6	·	·	8
HLA_MHCII	**5,449**	1	·	·	·	·
Homeostatic	**3,683**	9	2	·	4	·
Homeostatic_MERTK	**1,002**	3	·	·	·	·
IRM	**708**	2	·	·	·	·
Lipid_LAM	**1,784**	1	·	·	4	·

Author column: `cell\_type`

2.8. AlbertiniZielonka2025 / scrna_lecMicro_cleaned_integrated_sctransform

Tech: 10x_chromium, n_cells: 22,420 ⚠ 43 pending columns not merged

2.8.1. QA flags

Cells	22,420
Gene overlap	2,041 / 2,727
Median residual	1.972
WholeCell% (high)	47.4% ⚠ likely whole-cell
DC% (> 0.05)	53.7% ⚠ DC contamination

2.8.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	10,549	47.1%
Homeostatic	3,231	14.4%
Lipid_LAM	2,098	9.4%
HLA_MHCII	1,725	7.7%
IRM	1,602	7.1%
DC_pDC	1,043	4.7%
DC_cDC2	758	3.4%
Border_CAM	624	2.8%
Cycling	241	1.1%
Monocyte_Derived	203	0.9%
other (3)	346	1.5%

2.8.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM	DAM	IRM	HLA	CRM	RM
Homeostatic_MERTK	10,549	0.30	0.18	0.07	0.49	0.23	0.50
Homeostatic	3,231	0.26	0.22	0.09	0.54	0.24	0.53
Lipid_LAM	2,098	0.05	0.42	0.11	0.72	0.15	0.68
HLA_MHCII	1,725	0.14	0.28	0.12	0.75	0.22	0.71
IRM	1,602	0.16	0.25	0.28	0.61	0.22	0.60
DC_pDC	1,043	0.15	0.26	0.09	0.70	0.20	0.74
DC_cDC2	758	0.12	0.26	0.08	0.76	0.16	0.83
Border_CAM	624	0.21	0.23	0.09	0.61	0.25	0.62
Cycling	241	0.14	0.22	0.09	0.58	0.16	0.64
Monocyte_Derived	203	0.10	0.27	0.11	0.76	0.19	0.80

Author column mapping: HM→HM_enriched1, DAM→DAM_enriched1, IRM→IRM_enriched1, HLA→HLA_enriched1, CRM→tCRM_enriched1, RM→TRM_enriched1

2.8.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `Cluster\_coarse`. Concentration along rows = agreement.

starcat label	HM	RM	CRM	DAM	tCRM	IRM	HLA	C3-high	Prolif
Border_CAM	**177**	**103**	**144**	19	95	29	30	24	3
Cycling	21	15	22	1	16	6	·	4	**156**
DAM	41	16	41	12	23	7	1	4	1
DC_Mature	2	7	12	·	3	12	1	·	·
DC_cDC1	23	52	30	7	14	14	23	·	·
DC_cDC2	**132**	**306**	81	30	53	31	**111**	12	2
DC_pDC	**171**	**305**	**194**	49	**125**	71	95	23	10
HLA_MHCII	**216**	**268**	**296**	**413**	**202**	93	**203**	26	8
Homeostatic	**1,192**	**400**	**386**	**377**	**460**	**188**	57	**146**	25
Homeostatic_MERTK	**4,918**	**1,582**	**1,255**	**296**	**1,240**	**569**	**172**	**460**	57
IRM	64	81	**192**	36	79	**1,095**	41	13	1
Lipid_LAM	81	**245**	**253**	**1,344**	85	55	8	24	3

Author column: Cluster_coarse

2.9. Gabitto2024 / Seattle_MTG_MG

Tech: 10x_chromium, n_cells: 40,000 ⚠ 43 pending columns not merged

2.9.1. QA flags

Cells	40,000
Gene overlap	2,727 / 2,727
Median residual	1.090
WholeCell% (high)	0.8%
DC% (> 0.05)	64.7% ⚠ DC contamination

2.9.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	22,236	55.6%
Homeostatic_MERTK	6,244	15.6%
Lipid_LAM	2,937	7.3%
DAM	2,670	6.7%
Border_CAM	1,812	4.5%
DC_pDC	1,208	3.0%
HLA_MHCII	1,017	2.5%
Monocyte_Derived	666	1.7%
DC_cDC2	348	0.9%
IRM	346	0.9%
other (3)	516	1.3%

No comparable author state-scores or label columns detected in this dataset; starcat annotation stands on its own.

2.10. Simpson2025 / module_scored_sorl1

Tech: 10x_chromium, n_cells: 42,270 ⚠ 43 pending columns not merged

2.10.1. QA flags

Cells	42,270
Gene overlap	2,716 / 2,727
Median residual	2.974
WholeCell% (high)	18.9%
DC% (> 0.05)	44.7% ⚠ DC contamination

2.10.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	31,749	75.1%
Homeostatic	5,726	13.5%
Lipid_LAM	2,567	6.1%
HLA_MHCII	618	1.5%
IRM	607	1.4%
DC_cDC2	300	0.7%
DC_pDC	257	0.6%
Cycling	233	0.6%
Monocyte_Derived	115	0.3%
Border_CAM	71	0.2%
other (3)	27	0.1%

2.10.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM	DAM	IRM	HLA	CRM	RM	prolif	DC	macro
Homeostatic_MERTK	31,749	0.82	0.27	0.07	0.79	0.10	0.94	−0.07	−0.02	0.09
Homeostatic	5,726	0.73	0.33	0.09	0.80	0.15	0.81	−0.07	−0.01	−0.11
Lipid_LAM	2,567	0.44	0.75	0.10	1.04	0.15	1.27	0.06	−0.06	−0.13
HLA_MHCII	618	0.55	0.45	0.13	1.23	0.23	1.23	−0.01	0.04	−0.08
IRM	607	0.61	0.39	0.69	0.89	0.23	1.07	−0.05	−0.04	0.13
DC_cDC2	300	0.50	0.35	0.10	1.33	0.19	1.57	0.02	0.47	−0.06
DC_pDC	257	0.54	0.39	0.11	1.25	0.26	1.52	−0.00	0.40	−0.03
Cycling	233	0.60	0.28	0.09	0.73	0.07	0.82	0.70	−0.08	−0.04
Monocyte_Derived	115	0.47	0.33	0.14	1.23	0.38	1.24	0.01	0.82	−0.14
Border_CAM	71	0.59	0.36	0.12	0.93	0.21	1.13	0.09	0.17	0.40

Author column mapping: HM→HM8, DAM→DAM6, IRM→IRM9, HLA→HLA7, CRM→CRM5, RM→RM10, prolif→proliferating2, DC→dendritic1, macro→macrophage3

2.10.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `cluster\_id`. Concentration along rows = agreement.

starcat label	0	1	2	3	4	5	6	7
Border_CAM	28	4	·	12	11	3	2	11
Cycling	·	·	·	2	1	·	·	**230**
DAM	·	7	·	6	·	1	·	·
DC_Mature	·	·	·	2	·	·	·	·
DC_cDC1	·	3	·	·	4	1	1	2
DC_cDC2	8	21	1	14	**189**	1	64	2
DC_pDC	11	22	·	31	**143**	2	44	4
HLA_MHCII	27	**401**	12	60	81	7	27	3
Homeostatic	**894**	**2,036**	**986**	**794**	**615**	**282**	49	70
Homeostatic_MERTK	**12,215**	**3,288**	**6,354**	**5,485**	**1,756**	**1,451**	**804**	**396**
IRM	·	22	·	40	23	**506**	13	3
Lipid_LAM	13	**2,147**	2	**184**	33	33	**150**	5

Author column: `cluster\_id`

2.11. Simpson2025 / seurat_obj_sorl1_final

Tech: 10x_chromium, n_cells: 42,270 ⚠ 43 pending columns not merged

2.11.1. QA flags

Cells	42,270
Gene overlap	2,723 / 2,727
Median residual	1.663
WholeCell% (high)	16.8%
DC% (> 0.05)	56.4% ⚠ DC contamination

2.11.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	32,584	77.1%
Homeostatic	5,059	12.0%
Lipid_LAM	2,333	5.5%
IRM	555	1.3%
HLA_MHCII	510	1.2%
DC_cDC2	453	1.1%
DC_pDC	329	0.8%
Cycling	204	0.5%
Monocyte_Derived	101	0.2%
Border_CAM	97	0.2%
other (3)	45	0.1%

2.11.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM	DAM	IRM	HLA	CRM	RM	prolif	DC	macro
Homeostatic_MERTK	32,584	0.82	0.28	0.07	0.79	0.11	0.92	−0.07	−0.01	0.08
Homeostatic	5,059	0.71	0.34	0.09	0.81	0.16	0.82	−0.07	−0.01	−0.12
Lipid_LAM	2,333	0.43	0.77	0.10	1.04	0.14	1.27	0.06	−0.06	−0.13
IRM	555	0.60	0.39	0.70	0.89	0.24	1.07	−0.04	−0.04	0.13
HLA_MHCII	510	0.54	0.47	0.13	1.25	0.22	1.25	−0.01	0.01	−0.09
DC_cDC2	453	0.44	0.39	0.10	1.39	0.15	1.76	0.04	0.34	−0.04
DC_pDC	329	0.47	0.42	0.10	1.35	0.15	1.76	0.03	0.23	−0.01
Cycling	204	0.57	0.28	0.09	0.74	0.07	0.85	0.73	−0.07	−0.04
Monocyte_Derived	101	0.45	0.32	0.12	1.21	0.34	1.29	−0.00	0.74	−0.12
Border_CAM	97	0.54	0.37	0.10	1.00	0.17	1.29	0.07	0.16	0.32

Author column mapping: HM→HM8, DAM→DAM6, IRM→IRM9, HLA→HLA7, CRM→CRM5, RM→RM10, prolif→proliferating2, DC→dendritic1, macro→macrophage3

2.11.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `seurat\_clusters`. Concentration along rows = agreement.

starcat label	0	1	2	3	4	5	6	7
Border_CAM	31	5	5	12	14	4	15	11
Cycling	2	1	·	2	1	·	3	**195**
DAM	2	13	1	5	·	1	·	·
DC_Mature	·	·	·	2	·	·	1	·
DC_cDC1	1	·	1	1	4	·	13	·
DC_cDC2	21	39	3	21	**197**	2	**168**	2
DC_pDC	22	28	3	22	**102**	1	**148**	3
HLA_MHCII	21	**348**	8	50	53	5	22	3
Homeostatic	**744**	**1,899**	**826**	**699**	**518**	**261**	41	71
Homeostatic_MERTK	**12,335**	**3,626**	**6,505**	**5,636**	**1,949**	**1,526**	**577**	**430**
IRM	4	21	·	34	20	**456**	15	5
Lipid_LAM	9	**1,972**	2	**149**	22	30	**143**	6

Author column: `seurat\_clusters`

2.12. Gabitto2024 / Seattle_PFC_MG

Tech: 10x_chromium, n_cells: 42,486 ⚠ 43 pending columns not merged

2.12.1. QA flags

Cells	42,486
Gene overlap	2,727 / 2,727
Median residual	1.095
WholeCell% (high)	0.7%
DC% (> 0.05)	67.1% ⚠ DC contamination

2.12.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	24,208	57.0%
Homeostatic_MERTK	5,670	13.3%
Lipid_LAM	3,247	7.6%
DAM	2,715	6.4%
Border_CAM	2,152	5.1%
DC_pDC	1,366	3.2%
HLA_MHCII	1,095	2.6%
Monocyte_Derived	725	1.7%
IRM	358	0.8%
Cycling	357	0.8%
other (3)	593	1.4%

No comparable author state-scores or label columns detected in this dataset; starcat annotation stands on its own.

2.13. Davis2025 / 20240812_sexmicroglia

Tech: 10x_chromium, n_cells: 43,552 ⚠ 43 pending columns not merged

2.13.1. QA flags

Cells	43,552
Gene overlap	2,727 / 2,727
Median residual	1.333
WholeCell% (high)	25.0%
DC% (> 0.05)	49.9% ⚠ DC contamination

2.13.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	25,108	57.7%
Homeostatic	6,696	15.4%
HLA_MHCII	4,038	9.3%
Lipid_LAM	3,981	9.1%
IRM	1,609	3.7%
DC_pDC	1,089	2.5%
DC_cDC2	371	0.9%
Border_CAM	229	0.5%
Cycling	168	0.4%
DC_cDC1	138	0.3%
other (3)	125	0.3%

2.13.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM	DAM	IRM	HLA	CRM	RM	prolif
Homeostatic_MERTK	25,108	0.70	0.18	0.11	0.73	0.23	0.72	−0.04
Homeostatic	6,696	0.63	0.24	0.12	0.78	0.26	0.68	−0.01
HLA_MHCII	4,038	0.49	0.34	0.20	1.22	0.33	1.13	0.11
Lipid_LAM	3,981	0.38	0.58	0.16	1.11	0.29	1.04	0.22
IRM	1,609	0.46	0.31	0.55	1.03	0.41	0.99	0.07
DC_pDC	1,089	0.51	0.30	0.13	1.07	0.35	1.18	0.15
DC_cDC2	371	0.52	0.30	0.13	1.16	0.26	1.28	0.16
Border_CAM	229	0.50	0.27	0.14	0.90	0.28	0.98	0.08
Cycling	168	0.51	0.29	0.13	1.06	0.29	1.19	0.33
DC_cDC1	138	0.51	0.29	0.11	1.06	0.35	1.17	0.13

Author column mapping: HM→HM, DAM→DAM, IRM→IRM, HLA→HLA, CRM→CRM, RM→TRM, prolif→proliferating

2.13.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's annotations. Concentration along rows = agreement.

starcat label	HM	CRM	tCRM	HLA	RM	DAM	IRM
Border_CAM	46	28	97	5	21	17	15
Cycling	22	17	**101**	5	22	·	1
DAM	6	1	7	3	2	3	4
DC_Mature	3	3	26	·	3	·	4
DC_cDC1	17	10	87	5	12	1	6
DC_cDC2	82	36	**153**	17	69	7	7
DC_pDC	**109**	**162**	**619**	91	86	1	21
HLA_MHCII	**713**	**710**	**1,003**	**734**	**707**	82	89
Homeostatic	**3,041**	**1,730**	**458**	**856**	**365**	**127**	**119**
Homeostatic_MERTK	**10,882**	**7,584**	**3,912**	**1,639**	**610**	**125**	**356**
IRM	**114**	**446**	**282**	44	**112**	16	**595**
Lipid_LAM	**357**	**113**	**156**	**944**	**375**	**2,006**	30

Author column: annotations

2.14. Davis2025 / davis2025.20250619

Tech: 10x_chromium, n_cells: 43,552 ⚠ 43 pending columns not merged

2.14.1. QA flags

Cells	43,552
Gene overlap	2,727 / 2,727
Median residual	1.333
WholeCell% (high)	25.0%
DC% (> 0.05)	49.9% ⚠ DC contamination

2.14.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	25,108	57.7%
Homeostatic	6,696	15.4%
HLA_MHCII	4,038	9.3%
Lipid_LAM	3,981	9.1%
IRM	1,609	3.7%
DC_pDC	1,089	2.5%
DC_cDC2	371	0.9%
Border_CAM	229	0.5%
Cycling	168	0.4%
DC_cDC1	138	0.3%
other (3)	125	0.3%

2.14.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM	DAM	IRM	HLA	CRM	RM	prolif
Homeostatic_MERTK	25,108	0.70	0.18	0.11	0.73	0.23	0.72	−0.04
Homeostatic	6,696	0.63	0.24	0.12	0.78	0.26	0.68	−0.01
HLA_MHCII	4,038	0.49	0.34	0.20	1.22	0.33	1.13	0.11
Lipid_LAM	3,981	0.38	0.58	0.16	1.11	0.29	1.04	0.22
IRM	1,609	0.46	0.31	0.55	1.03	0.41	0.99	0.07
DC_pDC	1,089	0.51	0.30	0.13	1.07	0.35	1.18	0.15
DC_cDC2	371	0.52	0.30	0.13	1.16	0.26	1.28	0.16
Border_CAM	229	0.50	0.27	0.14	0.90	0.28	0.98	0.08
Cycling	168	0.51	0.29	0.13	1.06	0.29	1.19	0.33
DC_cDC1	138	0.51	0.29	0.11	1.06	0.35	1.17	0.13

Author column mapping: HM→HM, DAM→DAM, IRM→IRM, HLA→HLA, CRM→CRM, RM→TRM, prolif→proliferating

2.14.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's annotations. Concentration along rows = agreement.

starcat label	HM	CRM	tCRM	HLA	RM	DAM	IRM
Border_CAM	46	28	97	5	21	17	15
Cycling	22	17	**101**	5	22	·	1
DAM	6	1	7	3	2	3	4
DC_Mature	3	3	26	·	3	·	4
DC_cDC1	17	10	87	5	12	1	6
DC_cDC2	82	36	**153**	17	69	7	7
DC_pDC	**109**	**162**	**619**	91	86	1	21
HLA_MHCII	**713**	**710**	**1,003**	**734**	**707**	82	89
Homeostatic	**3,041**	**1,730**	**458**	**856**	**365**	**127**	**119**
Homeostatic_MERTK	**10,882**	**7,584**	**3,912**	**1,639**	**610**	**125**	**356**
IRM	**114**	**446**	**282**	44	**112**	16	**595**
Lipid_LAM	**357**	**113**	**156**	**944**	**375**	**2,006**	30

Author column: annotations

2.15. Gazestani2023 / microglia

Tech: 10x_chromium, n_cells: 67,513 ⚠ 43 pending columns not merged

2.15.1. QA flags

Cells	67,513
Gene overlap	2,727 / 2,727
Median residual	1.226
WholeCell% (high)	0.3%
DC% (> 0.05)	44.8% ⚠ DC contamination

2.15.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	39,952	59.2%
Homeostatic_MERTK	18,768	27.8%
Lipid_LAM	3,866	5.7%
Border_CAM	2,094	3.1%
HLA_MHCII	1,012	1.5%
DAM	801	1.2%
Monocyte_Derived	462	0.7%
DC_pDC	216	0.3%
IRM	143	0.2%
DC_cDC2	99	0.1%
other (3)	100	0.1%

2.15.3. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `anno\_orig\_state`. Concentration along rows = agreement.

Identity_Label	Microglia_PAM	Microglia_MGnD	Microglia_CCL3
Border_CAM	1	3	7	14	21	**217**	12	89	**1,531**	3
Cycling	1	1	1	·	6	1	·	·	·	1
DAM	11	6	70	**126**	81	**104**	64	**257**	7	5
DC_Mature	·	·	·	·	·	·	1	·	·	5
DC_cDC1	·	·	2	·	5	2	1	·	·	·
DC_cDC2	1	5	23	9	20	15	14	·	2	·
DC_pDC	3	3	14	8	44	17	6	2	1	1
HLA_MHCII	22	27	**159**	62	**528**	95	29	4	1	3
Homeostatic	**16,072**	**5,164**	**5,608**	**3,444**	**2,357**	**507**	**905**	75	4	**535**
Homeostatic_MERTK	**1,285**	**7,813**	**1,973**	**1,209**	**1,919**	**1,566**	**582**	34	19	**117**
IRM	·	2	13	10	31	28	13	3	1	·
Lipid_LAM	26	6	69	**1,295**	**282**	84	61	**1,174**	2	**307**

Author column: `anno\_orig\_state` (showing top 10 of 13 levels)

2.16. Blackburn2026 / seurat_obj_final

Tech: 10x_chromium, n_cells: 73,888 ⚠ 43 pending columns not merged

2.16.1. QA flags

Cells	73,888
Gene overlap	2,724 / 2,727
Median residual	2.408
WholeCell% (high)	17.8%
DC% (> 0.05)	61.1% ⚠ DC contamination

2.16.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	54,316	73.5%
Homeostatic	8,655	11.7%
Lipid_LAM	4,845	6.6%
HLA_MHCII	1,982	2.7%
IRM	1,187	1.6%
DC_pDC	1,116	1.5%
DC_cDC2	601	0.8%
Cycling	580	0.8%
Border_CAM	330	0.4%
Monocyte_Derived	203	0.3%
other (3)	73	0.1%

2.16.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM	DAM	IRM	HLA	CRM	RM	prolif	DC	macro
Homeostatic_MERTK	54,316	0.74	0.28	0.09	0.82	0.26	0.69	−0.05	−0.01	0.00
Homeostatic	8,655	0.65	0.35	0.09	0.83	0.31	0.63	−0.04	−0.08	−0.13
Lipid_LAM	4,845	0.35	0.77	0.09	1.08	0.32	1.04	0.07	−0.07	−0.17
HLA_MHCII	1,982	0.49	0.47	0.12	1.22	0.40	1.08	0.01	−0.04	−0.13
IRM	1,187	0.51	0.37	0.72	0.90	0.41	0.80	−0.03	−0.04	0.06
DC_pDC	1,116	0.47	0.39	0.10	1.17	0.40	1.19	0.03	0.19	−0.10
DC_cDC2	601	0.45	0.38	0.11	1.28	0.30	1.35	0.03	0.21	−0.11
Cycling	580	0.50	0.28	0.12	0.82	0.17	0.74	0.76	−0.05	−0.07
Border_CAM	330	0.49	0.38	0.15	0.99	0.38	0.93	0.06	0.10	0.36
Monocyte_Derived	203	0.36	0.37	0.13	1.34	0.42	1.34	0.03	0.39	−0.14

Author column mapping: HM→HM8, DAM→DAM6, IRM→IRM9, HLA→HLA7, CRM→CRM5, RM→RM10, prolif→proliferating2, DC→dendritic1, macro→macrophage3

2.16.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `seurat\_clusters`. Concentration along rows = agreement.

starcat label	0	1	2	3	4	5	6	7	8
Border_CAM	34	44	82	66	25	27	29	5	18
Cycling	·	·	1	·	2	1	**576**	·	·
DAM	4	6	3	31	5	1	·	1	·
DC_Mature	·	1	·	·	2	·	·	·	·
DC_cDC1	7	1	3	1	3	4	·	·	·
DC_cDC2	**117**	43	35	79	25	**284**	10	·	8
DC_pDC	**101**	**224**	**120**	70	40	**484**	55	·	22
HLA_MHCII	**136**	**255**	**213**	**1,103**	88	**125**	7	·	55
Homeostatic	**2,382**	**2,069**	**813**	**1,846**	**558**	**326**	**185**	**371**	**105**
Homeostatic_MERTK	**20,744**	**13,377**	**8,288**	**2,198**	**3,916**	**2,652**	**1,050**	**1,474**	**617**
IRM	7	39	38	**107**	**929**	49	12	1	5
Lipid_LAM	79	**180**	**507**	**3,868**	59	57	9	10	76

Author column: `seurat\_clusters`

2.17. Blackburn2026 / trisomy_seurat_obj_w_umap

Tech: 10x_chromium, n_cells: 73,888

2.17.1. QA flags

Cells	73,888
Gene overlap	2,700 / 2,727
Median residual	3.256
WholeCell% (high)	21.2%
DC% (> 0.05)	54.0% ⚠ DC contamination

2.17.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	52,810	71.5%
Homeostatic	8,762	11.9%
Lipid_LAM	5,418	7.3%
HLA_MHCII	2,452	3.3%
IRM	1,379	1.9%
DC_pDC	1,125	1.5%
Cycling	707	1.0%
DC_cDC2	601	0.8%
Border_CAM	294	0.4%
Monocyte_Derived	267	0.4%
other (3)	73	0.1%

2.17.3. Author state scores by starcat label

For each starcat-assigned Identity_Label, mean of the author's matching state-score columns. Diagonals (e.g. row=DAM, col=DAM) should be high.

Identity_Label	n	HM	DAM	IRM	HLA	CRM	RM	prolif	DC	macro
Homeostatic_MERTK	52,810	0.74	0.28	0.08	0.82	0.26	0.69	−0.05	−0.01	0.00
Homeostatic	8,762	0.66	0.34	0.09	0.82	0.30	0.62	−0.05	−0.08	−0.13
Lipid_LAM	5,418	0.37	0.76	0.09	1.07	0.32	1.03	0.06	−0.07	−0.17
HLA_MHCII	2,452	0.51	0.46	0.12	1.19	0.40	1.05	0.01	−0.04	−0.12
IRM	1,379	0.53	0.36	0.70	0.89	0.40	0.77	−0.03	−0.04	0.07
DC_pDC	1,125	0.49	0.38	0.10	1.15	0.41	1.16	0.02	0.20	−0.11
Cycling	707	0.53	0.29	0.12	0.81	0.18	0.72	0.72	−0.05	−0.07
DC_cDC2	601	0.48	0.37	0.11	1.25	0.32	1.29	0.02	0.22	−0.11
Border_CAM	294	0.51	0.39	0.14	1.00	0.39	0.93	0.06	0.10	0.39
Monocyte_Derived	267	0.37	0.37	0.13	1.36	0.42	1.37	0.02	0.44	−0.16

Author column mapping: HM→HM8, DAM→DAM6, IRM→IRM9, HLA→HLA7, CRM→CRM5, RM→RM10, prolif→proliferating2, DC→dendritic1, macro→macrophage3

2.17.4. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's seurat_clusters. Concentration along rows = agreement.

starcat label	0	1	2	3	4	5	6	7	8
Border_CAM	31	32	79	57	21	28	27	2	17
Cycling	·	1	3	·	1	1	**700**	·	1
DAM	2	7	4	25	5	·	2	·	1
DC_Mature	·	·	·	·	1	·	·	·	·
DC_cDC1	3	3	4	2	3	11	·	·	·
DC_cDC2	90	57	43	71	29	**296**	8	·	7
DC_pDC	91	**227**	98	47	34	**548**	55	·	25
HLA_MHCII	**186**	**350**	**281**	**1,277**	**118**	**165**	9	·	66
Homeostatic	**2,547**	**2,169**	**824**	**1,720**	**559**	**322**	**166**	**357**	98
Homeostatic_MERTK	**20,536**	**13,083**	**8,113**	**1,882**	**3,740**	**2,433**	**938**	**1,493**	**592**
IRM	7	67	45	**113**	**1,064**	62	15	·	6
Lipid_LAM	**123**	**242**	**607**	**4,178**	77	75	12	11	93

Author column: seurat_clusters

2.18. Borrie2025 / LOF.sc.teset.counts

Tech: 10x_chromium, n_cells: 77,684 ⚠ 43 pending columns not merged

2.18.1. QA flags

Cells	77,684
Gene overlap	2,058 / 2,727
Median residual	2.353
WholeCell% (high)	39.8% ⚠ likely whole-cell
DC% (> 0.05)	54.8% ⚠ DC contamination

2.18.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	40,891	52.6%
Homeostatic	12,580	16.2%
Lipid_LAM	5,729	7.4%
HLA_MHCII	5,142	6.6%
IRM	4,912	6.3%
DC_pDC	3,089	4.0%
DC_cDC2	1,785	2.3%
Border_CAM	1,732	2.2%
Cycling	587	0.8%
DAM	467	0.6%
other (3)	770	1.0%

No comparable author state-scores or label columns detected in this dataset; starcat annotation stands on its own.

2.19. GreenFujita2024 / microglia

Tech: 10x_chromium, n_cells: 86,612 ⚠ 43 pending columns not merged

2.19.1. QA flags

Cells	86,612
Gene overlap	2,727 / 2,727
Median residual	1.229
WholeCell% (high)	0.0%
DC% (> 0.05)	63.4% ⚠ DC contamination

2.19.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	54,864	63.3%
Homeostatic_MERTK	10,135	11.7%
DAM	5,309	6.1%
Border_CAM	4,892	5.6%
Lipid_LAM	4,710	5.4%
DC_pDC	2,677	3.1%
Monocyte_Derived	1,143	1.3%
Cycling	863	1.0%
DC_cDC1	666	0.8%
HLA_MHCII	545	0.6%
other (3)	808	0.9%

2.19.3. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `state`. Concentration along rows = agreement.

starcat label	Mic.2	Mic.7	Mic.5	Mic.3	Mic.6	Mic.8	Mic.4	Mic.12	Mic.10	Mic.9
Border_CAM	86	**338**	**419**	43	**688**	**107**	3	**943**	**129**	76
Cycling	3	2	5	·	11	3	·	·	12	·
DAM	55	**1,044**	**374**	32	**1,660**	**364**	11	**825**	**473**	54
DC_Mature	·	·	1	·	28	1	·	·	2	·
DC_cDC1	30	**157**	**139**	27	**154**	30	6	12	49	16
DC_cDC2	23	66	42	14	80	16	4	18	43	18
DC_pDC	**147**	**480**	**828**	**132**	**475**	**108**	19	70	**125**	79
HLA_MHCII	45	**103**	40	26	**132**	66	6	20	56	33
Homeostatic	**11,199**	**8,521**	**7,026**	**7,027**	**2,831**	**4,190**	**4,391**	**575**	**1,961**	**2,733**
Homeostatic_MERTK	**1,937**	**1,808**	**1,141**	**941**	**1,279**	**763**	**433**	**162**	**485**	**404**
IRM	12	25	25	16	**109**	6	·	26	46	15
Lipid_LAM	**179**	**284**	**388**	70	**314**	85	31	**1,211**	**192**	**104**

Author column: `state` (showing top 10 of 18 levels)

2.20. Blackburn2026 / seurat_obj_trisomy_initial

Tech: 10x_chromium, n_cells: 88,582 ⚠ 43 pending columns not merged

2.20.1. QA flags

Cells	88,582
Gene overlap	2,724 / 2,727
Median residual	2.647
WholeCell% (high)	14.2%
DC% (> 0.05)	72.1% ⚠ DC contamination

2.20.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	60,326	68.1%
Homeostatic	9,482	10.7%
Lipid_LAM	5,769	6.5%
Border_CAM	4,441	5.0%
HLA_MHCII	2,502	2.8%
DC_pDC	2,070	2.3%
IRM	1,551	1.8%
DC_cDC2	1,250	1.4%
Cycling	770	0.9%
Monocyte_Derived	272	0.3%
other (3)	149	0.2%

No comparable author state-scores or label columns detected in this dataset; starcat annotation stands on its own.

2.21. Mathys2024 / multiregion_microglia

Tech: 10x_chromium, n_cells: 92,800 ⚠ 43 pending columns not merged

2.21.1. QA flags

Cells	92,800
Gene overlap	2,727 / 2,727
Median residual	1.280
WholeCell% (high)	0.0%
DC% (> 0.05)	50.4% ⚠ DC contamination

2.21.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic	60,234	64.9%
Homeostatic_MERTK	12,921	13.9%
Lipid_LAM	5,683	6.1%
DAM	4,915	5.3%
Border_CAM	2,846	3.1%
DC_pDC	1,465	1.6%
HLA_MHCII	1,281	1.4%
Monocyte_Derived	1,024	1.1%
DC_cDC2	633	0.7%
DC_cDC1	595	0.6%
other (3)	1,203	1.3%

2.21.3. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `cell\_type\_high\_resolution`. Concentration along rows = agreement.

starcat label	Mic P2RY12	Mic TPT1	Mic DUSP1	CAMs	Mic MKI67	T cells
Border_CAM	**2,152**	**214**	**170**	**305**	2	3
Cycling	**235**	27	53	·	**238**	1
DAM	**4,123**	**322**	**317**	**153**	·	·
DC_Mature	31	6	59	·	·	·
DC_cDC1	**495**	42	57	1	·	·
DC_cDC2	**526**	46	55	5	1	·
DC_pDC	**1,216**	84	**153**	3	·	9
HLA_MHCII	**1,030**	**160**	83	8	·	·
Homeostatic	**55,207**	**3,154**	**1,751**	63	53	6
Homeostatic_MERTK	**11,696**	**892**	**309**	9	12	3
IRM	**442**	58	46	7	·	·
Lipid_LAM	**4,914**	**374**	**345**	50	·	·

Author column: `cell\_type\_high\_resolution`

2.22. Mancuso_2024 / Mancuso_2023

Tech: 10x_chromium, n_cells: 127,755 ⚠ 43 pending columns not merged

2.22.1. QA flags

Cells	127,755
Gene overlap	2,723 / 2,727
Median residual	1.954
WholeCell% (high)	56.0% ⚠ likely whole-cell
DC% (> 0.05)	60.8% ⚠ DC contamination

2.22.2. Identity_Label distribution

Identity_Label	n	%
Homeostatic_MERTK	60,091	47.0%
Homeostatic	18,332	14.3%
HLA_MHCII	13,181	10.3%
Lipid_LAM	10,224	8.0%
DC_pDC	7,151	5.6%
IRM	6,165	4.8%
DC_cDC2	6,057	4.7%
Monocyte_Derived	2,019	1.6%
Border_CAM	2,014	1.6%
DC_cDC1	1,004	0.8%
other (3)	1,517	1.2%

2.22.3. Author label cross-tab

Cell counts: rows = starcat Identity_Label, columns = author's `cell.state\_v2.1`. Concentration along rows = agreement.

starcat label	Homeostatic (HM)	Transitioning CRM	Disease-associated (DAM)	Translational response (TRM)	Cytokines response 2 (CRM-2)	Antigen-presenting response (HLA)	Interferon response (IRM)	Cytokines response 1 (CRM-1)
Border_CAM	**550**	**264**	**237**	**176**	**297**	**114**	**337**	39
Cycling	**276**	**169**	49	**102**	**161**	20	21	41
DAM	**119**	59	73	30	**106**	36	22	21
DC_Mature	31	25	25	5	95	11	1	19
DC_cDC1	**262**	**197**	**101**	**159**	**187**	47	11	40
DC_cDC2	**1,884**	**698**	**693**	**1,258**	**856**	**399**	87	**182**
DC_pDC	**2,345**	**1,218**	**609**	**1,134**	**1,091**	**285**	**145**	**324**
HLA_MHCII	**2,153**	**1,797**	**1,343**	**546**	**1,533**	**4,603**	**463**	**743**
Homeostatic	**7,340**	**2,923**	**2,308**	**1,345**	**1,661**	**1,377**	**544**	**834**
Homeostatic_MERTK	**34,978**	**8,740**	**1,694**	**7,336**	**2,974**	**1,490**	**1,553**	**1,315**
IRM	**353**	**386**	**400**	**118**	**1,034**	**183**	**3,470**	**221**
Lipid_LAM	**575**	**337**	**6,408**	**553**	**966**	**661**	**163**	**561**

Author column: `cell.state\_v2.1`

3. Caveats

1. **DAM vs Lipid_LAM naming.** The catalog's `Microglia_DAM` is the Keren-Shaul/Krasemann APOE^{+}CD163^{+} DAM; `Microglia_Lipid_LAM` is the Marschallinger SPP1^{+}GPNMB^{+} lipid-LAM. Many recent papers use "DAM" for what the catalog calls `Lipid_LAM`. Cross-tabs typically show high author-DAM in starcat rows labeled `Lipid_LAM` rather than `DAM` — this is correct, not a disagreement.
2. **High DC% in soft signal.** `DC_total > 0.05` triggers in 30–100 % of cells in most datasets, but the discrete `Identity_Label = DC_*` argmax is typically only a few percent. Use the discrete label, not the soft threshold, for population estimates.
3. **WholeCell_Baseline_high** is high (≥ 25 %) for whole-cell datasets, low (≤ 5 %) for snRNA-seq. Useful as a tech detector independent of the `.md` `technology` field.
4. **Residual** values reported here use the variance-normalised L2 formulation from `example/annotate.py`; the §7.5 threshold of 0.7 is from that same formulation. Most datasets score 1–3, with a few (`Olah2020` 4.1) showing poor catalog coverage — likely due to non-myeloid contamination.